

"ONE BATTERY. THREE LIVES. INFINITE IMPACT."

VOLTLOOP

"BUILT FOR INDONESIA EV GROWTH AND ISLAND ENERGY RESILIENCE."



M Y N FADLLILAH



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PROBLEM-SOLUTION FIT

THE BATTERY PARADOX

TRIPLE-LIFECYCLE BATTERY INFRASTRUCTURE

THE PROBLEM

COMING BATTERY TSUNAMI

- Indonesia targets 13M electric motorcycles EVs by 2030.
- IEA projects nearly 80 GWh of EV battery volumes reach end of first life globally by 2030 under STEPS
- second-life and EOL infrastructure is still nascent and fragmented

LOGISTICS ELECTRIFICATION BOTTLENECK

- 2 million liters of fuel burned daily for last-mile delivery
- Charging 3-4 hours = unacceptable downtime
- New battery cost Rp15M/motor = CAPEX shock

ISLAND ENERGY INJUSTICE

- 3.500+ villages stil rely on diesel generators
- Cost: Rp4.000-8.000/kWh (3-5x grid pricing)
- Children study in darkness, fishermen lose cold chain

THE SOLUTION

EV NEW BATTERY

Consumer mobility 5-8 years

B2B SWAP NETWORK

Logistics fleet - A 80% - 2-3 years

ISLAND MICROGRID

Off-grid energy-B60-80%-5-7 years

MATERIAL RECOVERY

Nickel, lithium, cobalt extraction

INTELLIGENCE LAYER

<3MIN BATTERY SWAP

-85% LANDFILL DIVERSION

-50% ENERGY COST

~100% CLOSED-LOOP RECOVERY



VALUE PROPOSITION - SEGMENTS - Key Partnership



VALUE PROPOSITIONS

For Fleet EV Operators:

- Monetize retired batteries (was liability)
- End-of-life regulatory compliance
- New carbon credit revenue stream

For Logistics Companies:

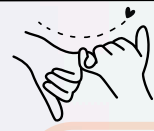
- Battery swap <3 min (vs 4-hour charge)
- TCO 30-40% cheaper than fuel
- Zero upfront capex (subscription)

For Off-Grid Communities:

- 24/7 electricity (vs diesel 4-6 hrs)
- 50% cheaper than diesel generators
- Clean, silent, reliable power

For OEM & Government:

- Real tropical degradation data
- Lifecycle traceability for policy



CUSTOMER RELATIONSHIPS

- **Logistics Companies:** Dedicated account manager and SLA (Service Level Agreement) contract.
- **Village Communities:** Cooperative partnership, local operator training, and monthly subscription.
- **EV Fleet Operators:** Long-term battery supply agreement.
- **OEMs/Government:** Data subscription, advisory services, and periodic reporting.



CHANNELS

- **Direct B2B Sales** to logistics companies and fleet operators.
- **Partnerships** with the government, PLN (State Electricity Company), and local governments.
- **Strategic Alliances** through the TBS Group network.
- **Village Cooperative Partnerships** for microgrid implementation.



CUSTOMER SEGMENTS

Primary Customers

- **B2B Logistics Companies:** J&T, SiCepat, JNE, Anteraja.
- **Village Electricity Cooperatives** and PLN Regional Business Units (PLN Wilayah Usaha) in 3T areas .

Strategic Customers

- **EV Fleet Operators:** TBS Group, ride-hailing fleets, and mining fleets.
- **Local Governments:** NTT (East Nusa Tenggara), Sulut (North Sulawesi), Maluku, and Papua.

Data Customers

- **Battery OEMs:** LG Energy, CATL, BYD.
- **Regulators:** ESDM (Ministry of Energy and Mineral Resources), KLHK (Ministry of Environment and Forestry).

KEY PARTNERS- KEY ACTIVITIES- KEY RESOURCES



KEY PARTNERS



KEY ACTIVITIES



KEY RESOURCES

Strategic Cooperation Partners

- • TBS Group: energy transition anchor, renewable energy access, waste management, ESG/circular economy pathway.
- • Electrum: EV motorcycle ecosystem, battery swapping infrastructure, fleet operation insight, and battery usage data.
- • GoTo/Gojek ecosystem: potential logistics/ride-hailing fleet pilot and demand validation.

Deployment Partners

- • PLN regional units, village electricity cooperatives, local governments, logistics fleet operators.
- Technology & Compliance Partners
- • Battery OEMs, BMS/IoT vendors, cloud providers, certified recyclers, and ESDM/KLHK compliance stakeholders.

Battery Lifecycle Operations

- Collect, sort, grade, and diagnose used EV batteries.
- Repackage safe batteries for swap hubs, fleets, and microgrids.
- Platform & Data Intelligence
- Build VoltOS for tracking, prediction, and compliance reporting.
- Build VoltSmart for SoH scoring and reuse decisions.

Pilot & Deployment Design

- Design swap, fleet, and microgrid pilot models.
- Validate sites, feasibility, unit economics, and workflows.
- Regulatory & Ecosystem Readiness
- Align battery reuse, safety, EOL, and circular economy compliance.

Technology Assets

- VoltSmart diagnostic system and VoltOS intelligence platform.
- SoH database, degradation model, and lifecycle traceability system.

Infrastructure & Operational Assets

- Swap hub and microgrid network blueprint.
- Battery testing SOP, safety grading, and reuse decision framework.
- Reverse logistics workflow for used EV batteries.

Human, Academic & Team Assets

- Technical team across battery technology, energy systems, software engineering, and field operations.
- Access to ITB faculty expertise and relevant laboratories for advisory, prototyping, and technical validation.

Strategic Data Moat

- Tropical battery degradation data as VoltLoop's long-term unfair advantage.

BUSINESS MODEL CANVAS

REVENUE STREAMS-COST STRUCTURES

Cost Category	Cost Driver / Model	Y1 Cost	Y5 Cost
Hub Operational COGS	Active hubs × Rp273,6M/hub/year	Rp547,20M	Rp21,89B
Microgrid Operational COGS	Active microgrids × Rp66M/unit/year	Rp66,00M	Rp3,96B
Battery Replacement Reserve	Replacement reserve for deployed batteries	Rp0	Rp7,40B
Warranty Reserve + Spares	Warranty reserve + active hub spare allowance	Rp15,29M	Rp1,76B
Total COGS	Operational COGS + battery reserve + warranty/spares	Rp628,49M	Rp35,00B
Sales & Marketing	30% of revenue in Y1 → 10% in Y5	Rp135,52M	Rp6,39B
R&D	40% of revenue in Y1 → 8% in Y5	Rp180,69M	Rp5,12B
G&A	40% of revenue in Y1 → 8% in Y5	Rp180,69M	Rp5,12B
EBITDA	Revenue – COGS – OPEX	–Rp673,67M	Rp18,98B

Asset / Unit	Base Cost Structure
Swap Hub	Capex ±Rp800M/hub
Microgrid Unit	Capex ±Rp280M/unit
Smart Battery	Mass unit cost ±Rp6,94M/pack
VoltSwap Station	Mass unit cost ±Rp37,54M/station
VoltGrid Cabinet	Mass unit cost ±Rp48,98M/unit

Revenue Stream	Pricing Model / Driver	Y1 Revenue	Y5 Revenue
Swap Subscription	Rp500.000/motor/month; grows to ±Rp573.762 by Y5	Rp90,00M	Rp18,18B
Microgrid Subscription	Rp150.000/household/month; grows to ±Rp172.128 by Y5	Rp282,24M	Rp21,25B
Battery-as-a-Service	Rp700.000–Rp900.000/customer/month uplift	Rp21,00M	Rp9,50B
Carbon Credit Monetization	Rp96.000/tCO ₂ × 70% monetization rate	Rp6,48M	Rp602,28M
Battery Passport SaaS	Rp2.400.000/battery/year; platform adoption 30% → 95%	Rp36,00M	Rp9,12B
OEM Analytics API	Rp400M–Rp459M/OEM/year; 0 → 10 OEM customers	Rp0	Rp4,59B
Recycling Revenue	Material recovery revenue starting Y3	Rp0	Rp360,00M
Other Revenue	Govt EPC / ESG consulting / project-based revenue	Rp16,00M	Rp340,00M
TOTAL REVENUE	Multi-stream infrastructure + data monetization	Rp451,72M	Rp63,95B

INFRASTRUCTURE-HEAVY UPFRONT; LONG-TERM MARGIN IMPROVES THROUGH HIGHER UTILIZATION, RECURRING REVENUE, AND SAAS/DATA MONETIZATION.

MARKET & COMPETITIVE LANDSCAPE

MARKET SIZE / COMPETITIVE POSITIONING / SUSTAINABILITY IMPACT

MARKET SIZE - INDONESIA TAM

Segment	TAM 2030 (Rp)
Battery swap for logistics	Rp4.80T
Off-grid microgrid energy	Rp1.73T
Battery lifecycle data services	Rp720B
Carbon credit (battery + microgrid)	Rp480B
TOTAL TAM	Rp7.75 T
TOTAL SAM	Rp2.71 T
TOTAL SOM	Rp29.01B

COMPETITIVE LANDSCAPE

PLAYER	B2C	B2B	2nd	Off	Data	iD
Gogoro (Taiwan)	Y	N	N	N	-	P
Swap Energy (ID)	Y	-	N	N	N	Y
Connected Energy (UK)	N	N	Y	N	N	N
Sunseap (SG)	N	N	N	Y	N	P
Tesla Energy (US)	N	N	N	N	Y	N
Electrum	Y	N	N	N	N	Y
VOLTLOOP	Y	Y	Y	Y	Y	Y

"CANDIDATE FIRST INTEGRATED CIRCULAR BATTERY PLATFORM FOCUSED ON INDONESIA ARCHIPELAGO USE CASE."

HARDWARE NETWORK

80 swap hubs + 60 microgrids by Y5

DATA MOAT

4,000 batteries on platform + 12 yr tropical degradation data

VERTICAL INTEGRATION

Fleet → swap → microgrid → passport → recovery

TBS STRATEGIC ANCHOR

Target LOI for 200 second-life batteries in Y1

REGULATORY FIRST-MOVER

Battery passport + circular compliance readiness

SUSTAINABILITY IMPACT (YEAR 5)

8,962 TONS CO₂ AVOIDED/YEAR | 51.7M L DIESEL DISPLACED | 10,500 HOUSEHOLDS ELECTRIFIED | 70 TONS BATTERY WASTE AVOIDED | 195 GREEN JOBS | SDG 7,9,11,12,13 ALIGNED

EXECUTION ROADMAP & FINANCIALS

5-YEAR PHASED DEPLOYMENT / FINANCIAL TRAJECTORY / KEY MILESTONES

VALIDATION	FIRST PILOT	EARLY SCALE	NATIONAL	ASEAN
Q1-Q2 2026 / 0-6 months - TBS/ELECTRUM SECOND-LIFE - J&T / SICEPAT 50-COURIER PILOT LOI - GRANT TARGET USD 200K–500K	Q3 2026 - Q2 2027 2 Swap Hubs Jakarta 1 Microgrid pilot - VoltOS v1 + battery passport beta	Q3 2027 - 2028 8 → 25 Swap Hubs 4 → 18 Microgrids - Series A Rp20B	2028 - 2030 50 → 80 Swap Hubs 38 → 60 Microgrids 4,400 couriers + 10,500 households	2030+ / Y5+ - Battery Passport ASEAN - OEM analytics/API expansion - Strategic exit / IPO

METRIC	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
Revenue	Rp451,72 jt	Rp2,98 M	Rp14,12 M	Rp34,79 M	Rp63,95 M
COGS	Rp 628.49 jt	Rp 2.71 M	Rp 9.27 M	Rp 20.18 M	Rp 35 M
Gross Profit	(Rp176,77 jt)	Rp268,79 jt	Rp4,84 M	Rp14,61 M	Rp28,95 M
Operating Expenses	Rp496,90 jt	Rp938,32 jt	Rp3,83 M	Rp7,17 M	Rp9,97 M
EBITDA	(Rp673,67 jt)	(Rp669,53 jt)	Rp1,01 M	Rp7,44 M	Rp18,98 M
Net Margin	—	—	15%	26%	36%

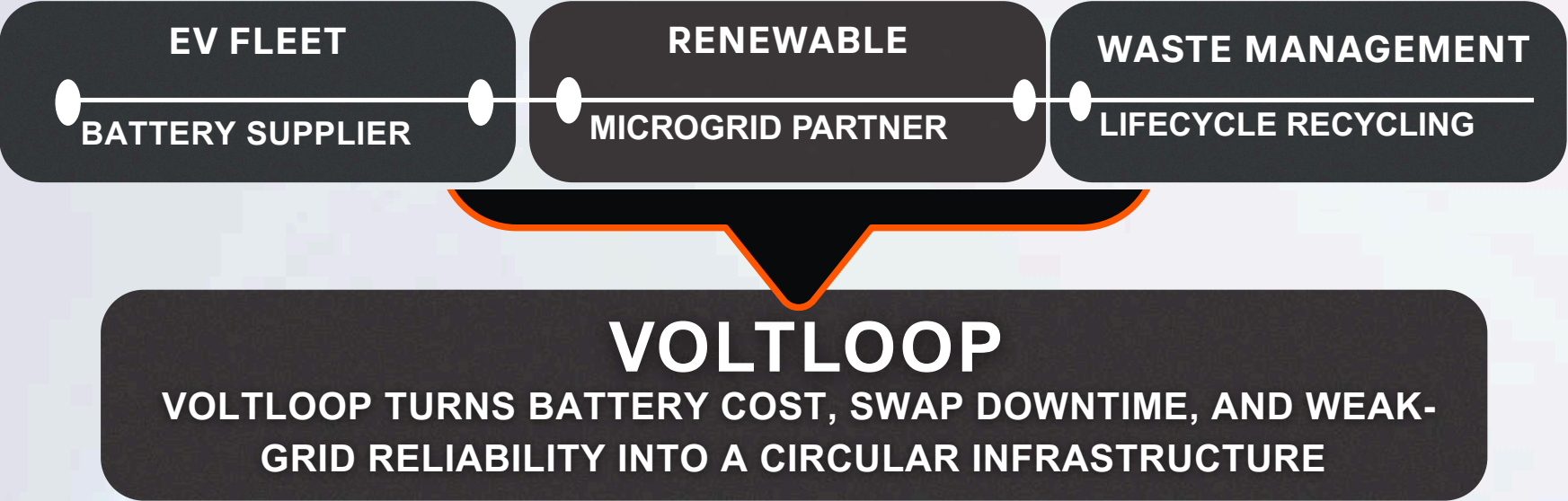


BUSINESS VIABILITY & EXECUTION

STRATEGIC FIT, RISK, & TEAM

TBS SYNERGY/ RISK MITIGATION / THE TEAM

STRATEGIC FIT WITH TBS GROUP



RISK ASSESSMENT & MITIGATION

RISK	PROB	IMPACT	MITIGATION	KPI
Battery economics not attractive	Med	Very High	Anchor fleet, BaaS pricing, asset financing	Driver TCO lower than ICE / current EV rental
Swap reliability / downtime	Med	High	Pre-charged buffer, VoltOS forecasting, uptime SLA	Hub uptime ≥95%; ready-pack availability
Battery safety / degradation	Low-Med	Very High	Layered BMS, SoH cutoff, lab testing, insurance	Zero critical incidents
Capex overrun	Med	High	Stage-gated deployment, no 25-hub rollout before KPI proof	Runway >12 months
Regulatory / PLN pathway	Med	High	Partner through TBS/PLN/authorized site owner;	Permit before capex
Partner concentration	Med	High	TBS as anchor, not sole dependency; multi-source	No single supplier >50% after pilot

AXION TEAM

Role	Responsibility
CEO / BD	TBS, Electrum, logistics anchor, fundraising
CTO / Lead Battery Engineer	BMS, SoH model, safety architecture, Battery Passport
COO	Hub deployment, fleet operations, SLA, field maintenance
CFO	Unit economics, grants, asset financing, runway control
Advisors	ITB battery/energy lab, regulatory advisor, TBS/Electrum mentors

THE ASK



STRATEGIC FIT IS STRONG, BUT EXECUTION MUST BE DISCIPLINED: USE TBS/ELECTRUM AS THE VALIDATION ANCHOR, PROVE UNIT ECONOMICS THROUGH A SMALL FLEET PILOT, THEN SCALE ONLY AFTER SAFETY, UPTIME, UTILIZATION, AND CASH RUNWAY KPIS ARE MET.



THANK
YOU

"ONE BATTERY. THREE LIVES. INFINITE IMPACT."

LAMPIRAN ∞

PROTOTYPE(ENG.DESIGN).

PROTO-VOLTOS

POC(PROOF OF CONCEPT)-VOLTOS

VIDEO--VOLTGRID-VOLTBATTERY

WORKBOOK

DATA SCRAPING

VIDEO PROMOSI